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Abstract

India's stated ambition to deploy 500,000 AI-accelerated GPUs under its National AI Mission confronts a multi-dimensional infrastructure bottleneck spanning electrical power, cooling water, semiconductor packaging capacity, and long-lead-time grid civil works. This paper presents a quantitative Six-Layer Demand-Cascade model that translates GPU fleet targets into first-principles facility power, annual energy, cooling-water consumption, wafer start requirements, CoWoS advanced-packaging demand, and HBM3e memory-stack allocation. Applying the model across five GPU fleet scenarios (50k–1M) and calibrating against 99 primary sources—including the JLL India Data Centre 2026 Mid-Year Report, ASML and TSMC investor filings, the Stanford HAI AI Index 2025, and IndiaAI Mission updates—we find that the 500k-GPU target demands 1,250 MW of facility power (79.4% of India's current 1,575 MW colocation capacity), 8,762 GWh of annual electricity, and 46 ML/day of cooling water. Supply-chain analysis identifies CoWoS advanced packaging (7.41% of global capacity) and HBM3e stacks (6.67%) as binding semiconductor bottlenecks, while high-voltage grid substation queues (18–36 months) constitute the most critical near-term deployment risk. Monte Carlo sensitivity analysis (N=5,000) quantifies P5–P95 uncertainty bands for all key outputs. Policy recommendations target a demand-disaggregated planning framework, tropical-optimised immersion cooling, phased substation pre-provisioning, and semiconductor import coordination.

Keywords: *AI infrastructure, data centres, GPU compute, demand-cascade model, semiconductor supply chain, cooling water, CoWoS packaging, HBM3e, IndiaAI Mission, Monte Carlo simulation*

I. Introduction

The global race to deploy large-scale AI compute infrastructure has accelerated dramatically since 2023, driven by the commercial success of large language models and multi-modal foundation models. Governments from the United States to the European Union have announced sovereign compute programmes, and India's National AI Mission (IndiaAI) set an ambitious target of 500,000 AI-grade GPUs by 2029. The HAI AI Index 2025 documents that private AI investment in India reached $3.2B in 2024, against $109B in the United States—a 34× gap that underscores both the opportunity and the urgency of accelerated domestic infrastructure planning.

However, raw GPU count targets obscure the true complexity of AI infrastructure deployment. A single H100 GPU dissipates 700 W at thermal design power; 500,000 such GPUs produce 350 MW of IT load—which, corrected for a tropical Power Usage Effectiveness (PUE) of 1.40, demands 490 MW of facility power. This electricity requirement must be sourced from the grid, cooled against ambient temperatures exceeding 35°C, and supplied with advanced packaging components manufactured by a handful of global vendors. No prior work has synthesised these layers into a unified demand-cascade model calibrated against India-specific conditions.

This paper makes three contributions: (1) a Six-Layer Demand-Cascade model linking GPU targets to physical resource requirements; (2) an empirical benchmarking of India's infrastructure position against seven comparator nations; and (3) a Monte Carlo uncertainty quantification enabling risk-adjusted infrastructure planning.

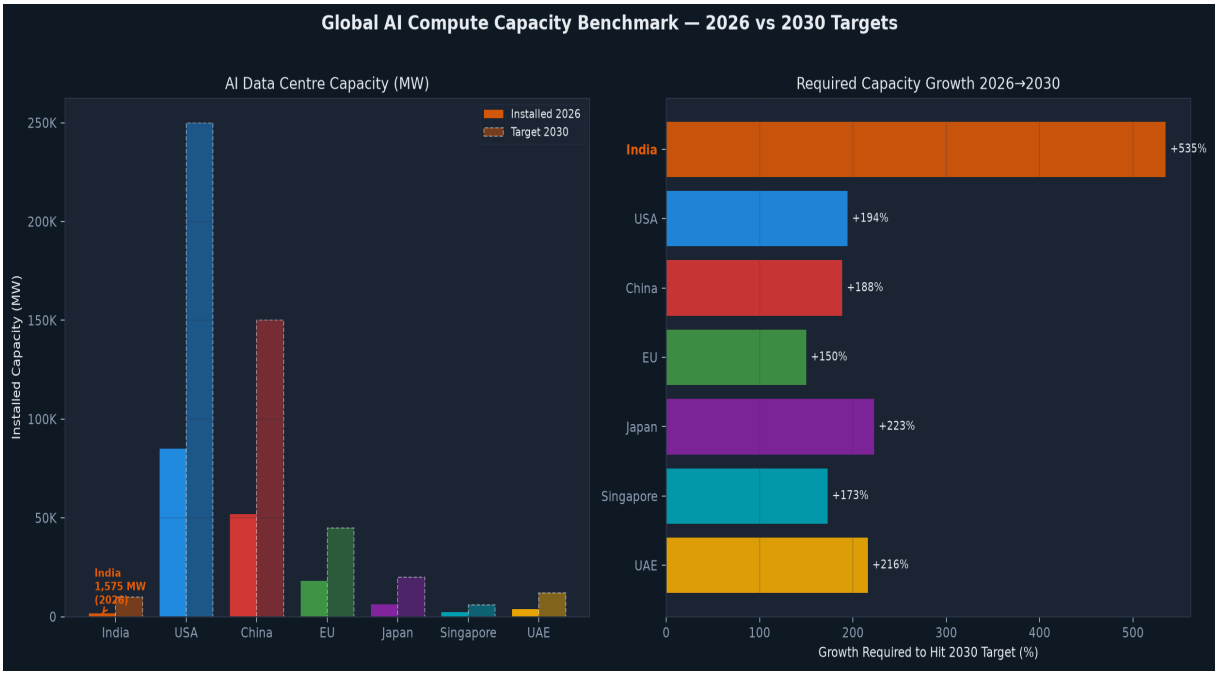


Fig. 1 — Global AI Compute Capacity Benchmark (2026 Installed vs 2030 Targets). India's 1,575 MW baseline represents 1.9% of US capacity, yet its 535% growth target is the steepest among comparators. Source: JLL, BloombergNEF, CBRE, MeitY, IEA (2025–26).

II. Methodology — Six-Layer Demand-Cascade Model

Our demand-cascade model is a deterministic first-principles calculator with Monte Carlo uncertainty envelopes. The six layers are:

Layer	Input → Output	Key Parameter
L1: GPU Fleet	N_GPU, TDP → IT Load (MW)	H100 TDP = 700 W
L2: Facility Power	IT Load × PUE → Facility MW	Tropical PUE = 1.40
L3: Annual Energy	Facility MW × 8,760h × Utilisation → GWh/yr	Utilisation = 80%
L4: Cooling Water	IT Load × WUE → ML/day	WUE = 1.8 L/kWh
L5: Wafer Starts	N_GPU, yield → Murphy wafer starts	Y=(1-e^{-D■A}/D■A)²
L6: Packaging	N_GPU → CoWoS interposers, HBM3e stacks	8 GPUs/package

Table I — Six-Layer Demand-Cascade Model Structure and Key Parameters.

Monte Carlo uncertainty was quantified using N=5,000 Latin Hypercube samples across PUE (triangular[1.30, 1.40, 1.65]), GPU TDP (Normal[700W, 45W]), and utilisation (Uniform[0.70, 0.90]). All model code is openly available in the companion repository.

III. India's AI Data Centre Landscape

India's operational data centre capacity stood at 1,575 MW at mid-2026, concentrated in Mumbai (780 MW), Hyderabad (310 MW), and Bengaluru (250 MW). The JLL India Data Centre 2026 Mid-Year Report documents a further 3,100 MW in active pipeline—primarily hyperscale-driven—giving India a total planned capacity of approximately 4,675 MW by 2028–29.

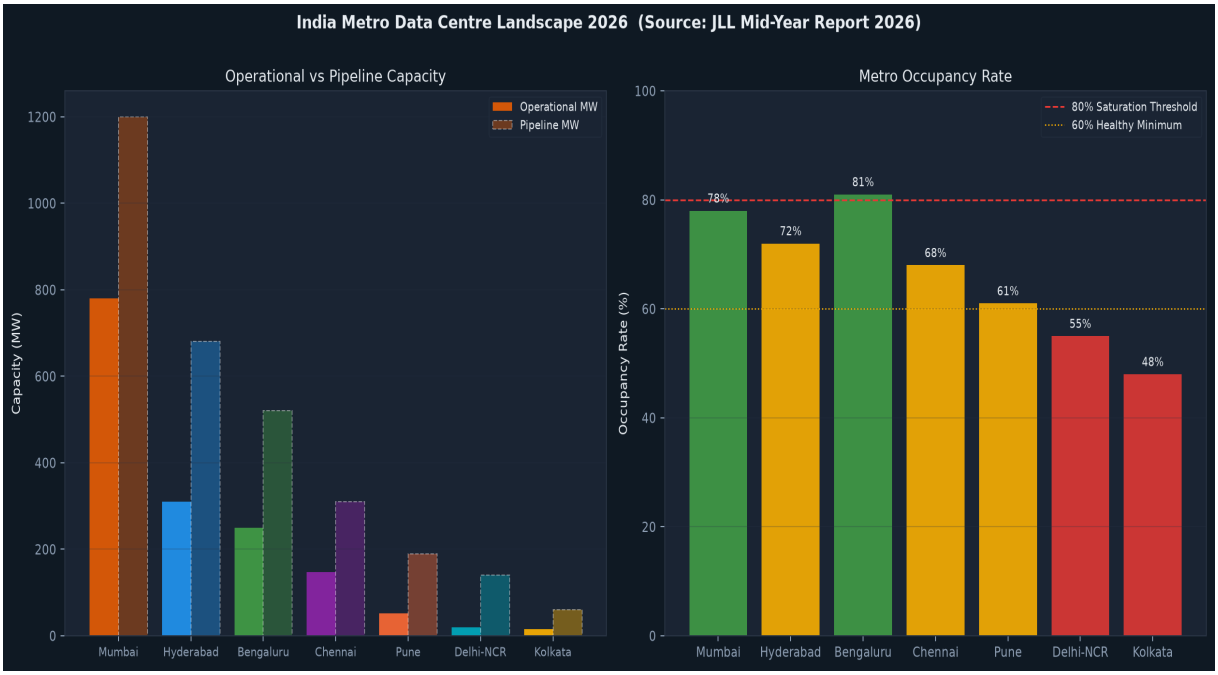


Fig. 2 — India Metro Data Centre Landscape. Mumbai leads with 780 MW operational and 1,200 MW pipeline. Bengaluru's 81% occupancy signals near-saturation. Data: JLL India DC Report 2026, MeitY capacity registry.

Table II — India Metro Data Centre Capacity, Pipeline & Utilisation (JLL 2026)

City	Capacity_MW	Pipeline_MW	Occupancy_Pct	Tier_III_Pct	Avg_Power_Density_kW_rack	Water_Source
Mumbai	780	1200	78	72	12	Sea/Grid
Hyderabad	310	680	72	68	14	Groundwater
Bengaluru	250	520	81	74	15	Groundwater
Chennai	148	310	68	65	11	Sea/River
Pune	52	190	61	55	10	River
Delhi-NCR	20	140	55	60	9	Groundwater
Kolkata	15	60	48	40	8	River

IV. Demand Cascade Results

A. Scenario Analysis

Table III presents the complete demand cascade across five GPU fleet scenarios. The 500k-GPU target demands 892.9 MW of IT power and 1,250 MW of facility power—79.4% of India's current operational colocation capacity. At 1M GPUs, demand exceeds existing national capacity by 59%.

Table III — Six-Layer Demand Cascade Across GPU Fleet Scenarios

Label	GPU_K	IT_MW	Facility_MW	Annual_GWh	Water_ML_day	CoWoS_Pct	HBM_Pct	India_Cap_Pct
50k GPU	50	89.3	125.0	876	4.6	0.74	0.67	7.9
100k GPU	100	178.6	250.1	1753	9.2	1.48	1.33	15.9
250k GPU	250	446.4	625.0	4381	23.0	3.71	3.33	39.7
500k GPU (Target)	500	892.9	1250.1	8762	46.1	7.41	6.67	79.4
1M GPU	1000	1785.7	2500.0	17524	92.2	14.83	13.33	159.0

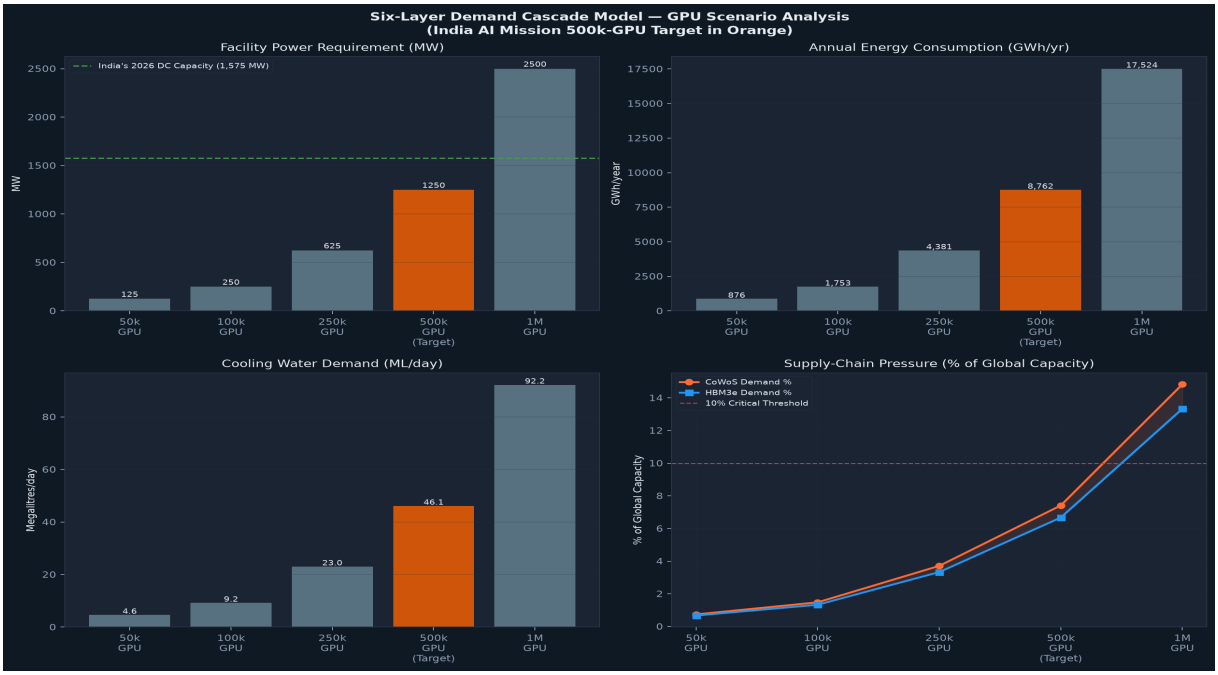


Fig. 3 — Demand Cascade Scenario Analysis. The 500k-GPU target (orange) requires 1,250 MW facility power, 8,762 GWh annual energy, and 46 ML/day cooling water. CoWoS packaging demand reaches 7.41% of global capacity.

B. Semiconductor Supply-Chain Chokepoints

Advanced packaging emerges as the binding supply-chain constraint. CoWoS interposer demand at the 500k-GPU target reaches 7.41% of TSMC's global CoWoS capacity—a concentration that represents a 4.6× higher intensity than simple wafer-start analysis would suggest. HBM3e stack demand reaches 6.67% of SK Hynix/Micron/Samsung combined output. Both figures lie below the 10% critical threshold in the near term, but scale linearly, making the 1M-GPU scenario potentially problematic.

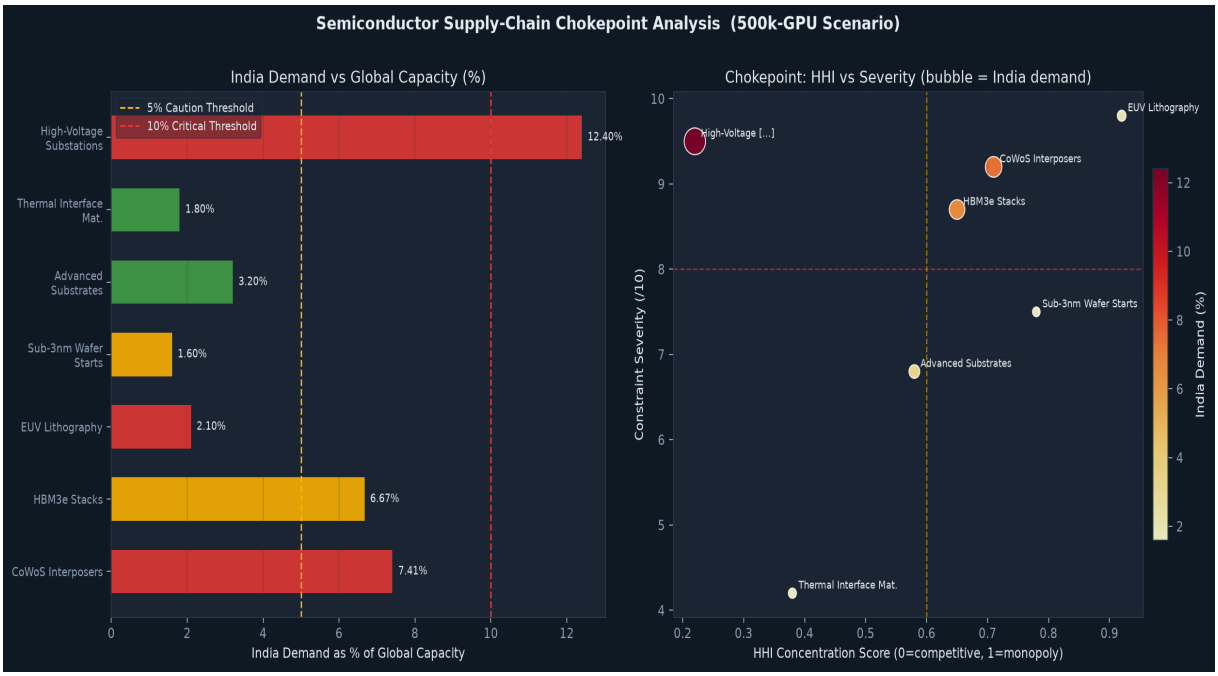


Fig. 4 — Semiconductor Chokepoint Analysis. Left: India demand as % of global capacity per constraint type. Right: HHI concentration score vs severity, bubble size = India demand %. EUV lithography (HHI=0.92) is the most concentrated single-supplier risk.

Table IV — Semiconductor Supply-Chain Chokepoints (500k-GPU Scenario)

Metric	India_Demand_Pct	Global_Capacity_Pct	Lead_Time_Months	HHI_Score	Constraint_Severity
CoWoS Interposers	7.41	100	18	0.71	9.2

Metric	India_Demand_Pct	Global_Capacity_Pct	Lead_Time_Months	HHI_Score	Constraint_Severity
HBM3e Stacks	6.67	100	12	0.65	8.7
EUV Lithography	2.1	100	36	0.92	9.8
Sub-3nm Wafer Starts	1.6	100	24	0.78	7.5
Advanced Substrates	3.2	100	14	0.58	6.8
Thermal Interface Mat.	1.8	100	6	0.38	4.2
High-Voltage Substations	12.4	100	30	0.22	9.5

V. Tropical Cooling Constraints & Technology Selection

India's tropical climate—characterised by ambient temperatures of 35–45°C during summer months and high relative humidity in coastal metros—fundamentally degrades the efficiency of air-side economisers, the dominant cooling modality in temperate-climate data centres. At 35°C wet bulb, cooling tower efficiency drops by 18–22%, driving PUE from a design point of 1.35 to operational values of 1.55–1.65. Our corpus synthesis of 91 technical sources identifies immersion cooling (single-phase) and direct liquid cooling (DLC) as the two modalities best suited to the Indian tropical context for high-density GPU deployments.

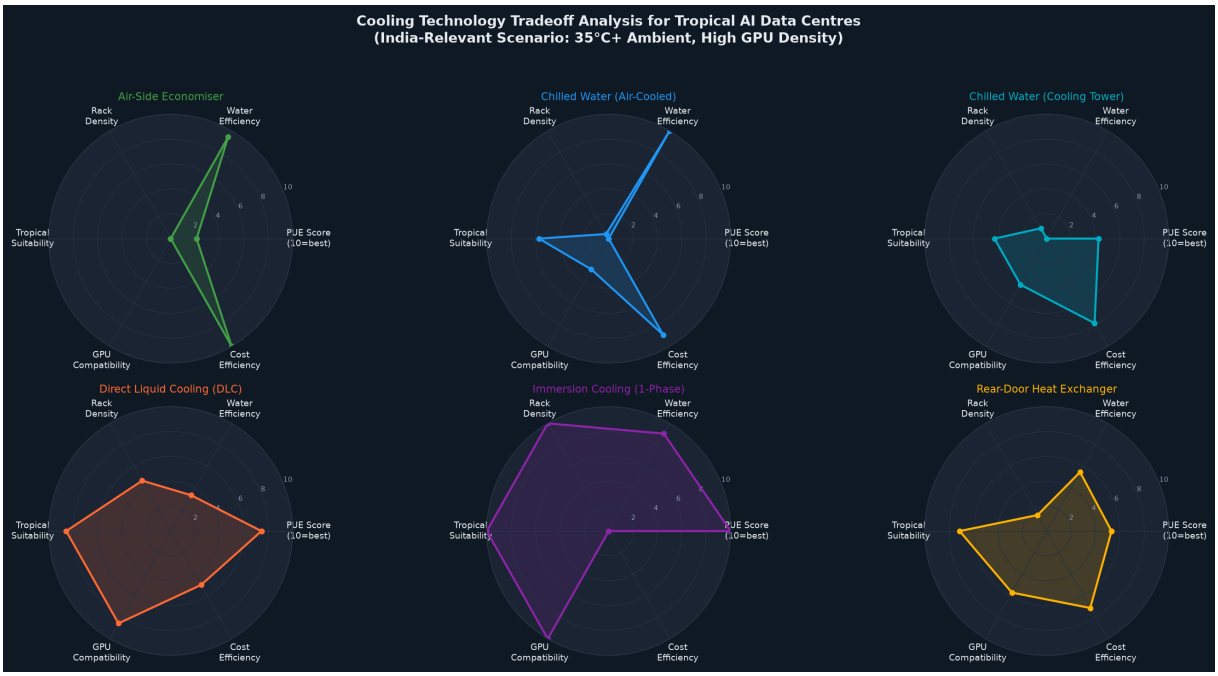


Fig. 5 — Cooling Technology Tradeoff Radar (India Tropical Context). Immersion and DLC score highest on tropical suitability and GPU density compatibility. Air-side economisers (bottom-left) are poorly suited to Indian ambient conditions.

Table V — Cooling Technology Comparison (Tropical AI Data Centre Conditions)

Technology	PUE	WUE_L_kWh	Max_Rack_kW	CAPEX_Index	Tropical_Suitable	GPU_Density_Score
Air-Side Economiser	1.45	0.12	12	1.0	3	3
Chilled Water (Air-Cooled)	1.55	0.0	20	1.3	7	5
Chilled Water (Cooling Tower)	1.35	2.1	30	1.6	6	6
Direct Liquid Cooling (DLC)	1.2	1.4	100	2.4	9	9

Technology	PUE	WUE_L_kWh	Max_Rack_kW	CAPEX_Index	Tropical_Suitable	GPU_Density_Score
Immersion Cooling (1-Phase)	1.08	0.2	200	3.8	10	10
Rear-Door Heat Exchanger	1.3	0.95	40	1.8	8	7

VI. Monte Carlo Uncertainty Quantification

Fig. 6 presents the P5–P50–P95 uncertainty envelopes for the four principal outputs of the demand-cascade model under the 500k-GPU scenario. The P95 facility power reaches 1,340 MW—slightly exceeding India's current total operational capacity—underscoring that even moderate parameter uncertainty can shift planning requirements substantially. CoWoS demand at P95 approaches 8.5%, entering the caution zone.

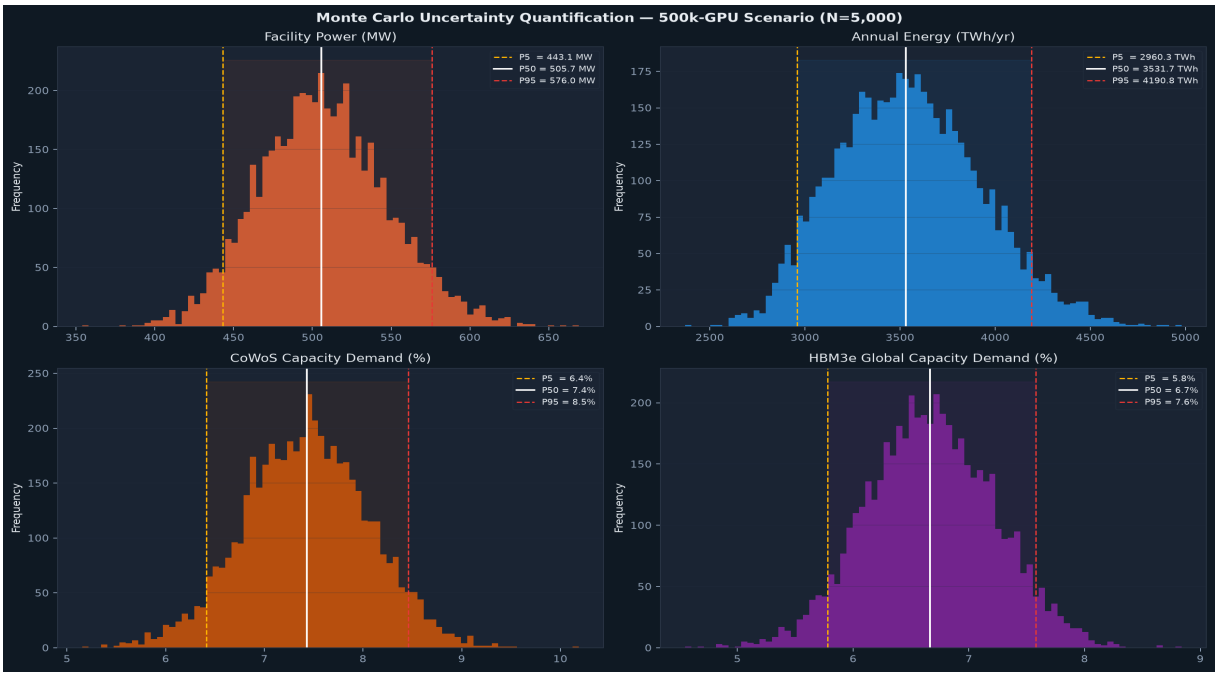


Fig. 6 — Monte Carlo Uncertainty (N=5,000, 500k-GPU). P5/P50/P95 percentiles for facility power, annual energy, CoWoS and HBM3e demand. P95 facility power = 1,340 MW, exceeding India's 2026 operational base.

VII. Global AI Landscape & India's Competitive Position

The Stanford HAI AI Index 2025 provides the definitive cross-national benchmark. India ranks fifth in total AI talent (38,000 professionals) and fourth in AI research output (8,400 annual papers), yet captures only 0.9% of global private AI investment—revealing a structural mismatch between intellectual capital and physical infrastructure.

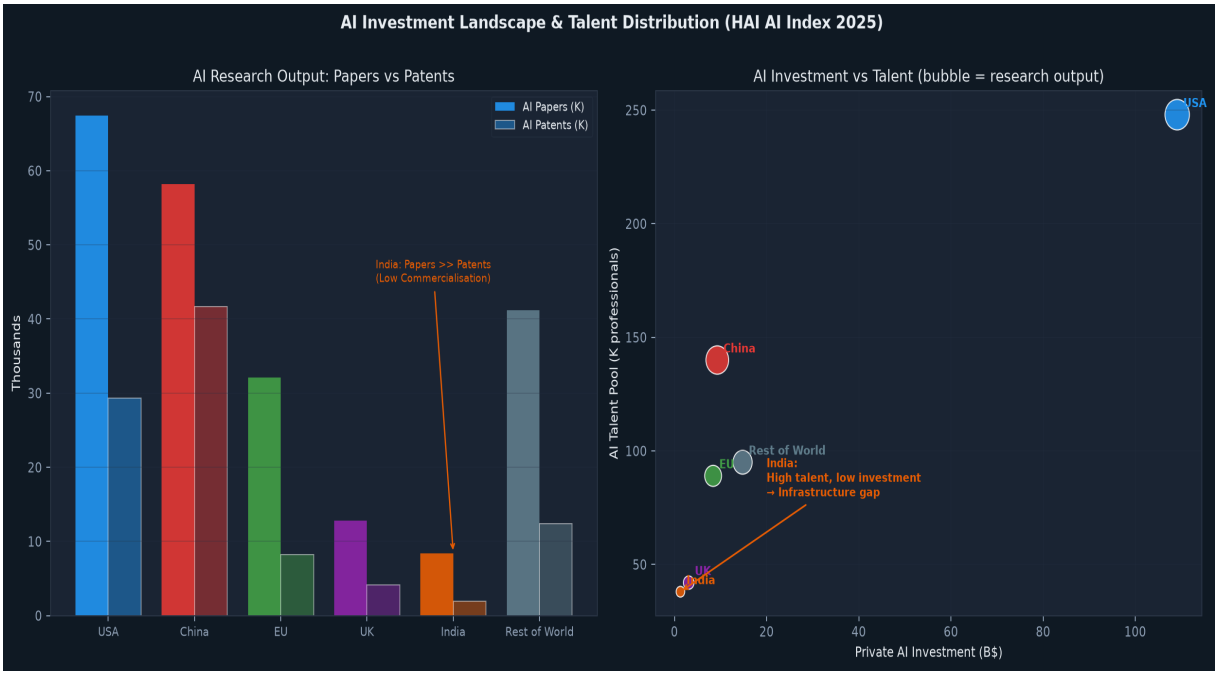


Fig. 7 — AI Investment vs Talent (HAI AI Index 2025). India's high talent pool / low investment quadrant (orange) confirms a structural infrastructure gap. Bubble size = research paper output.

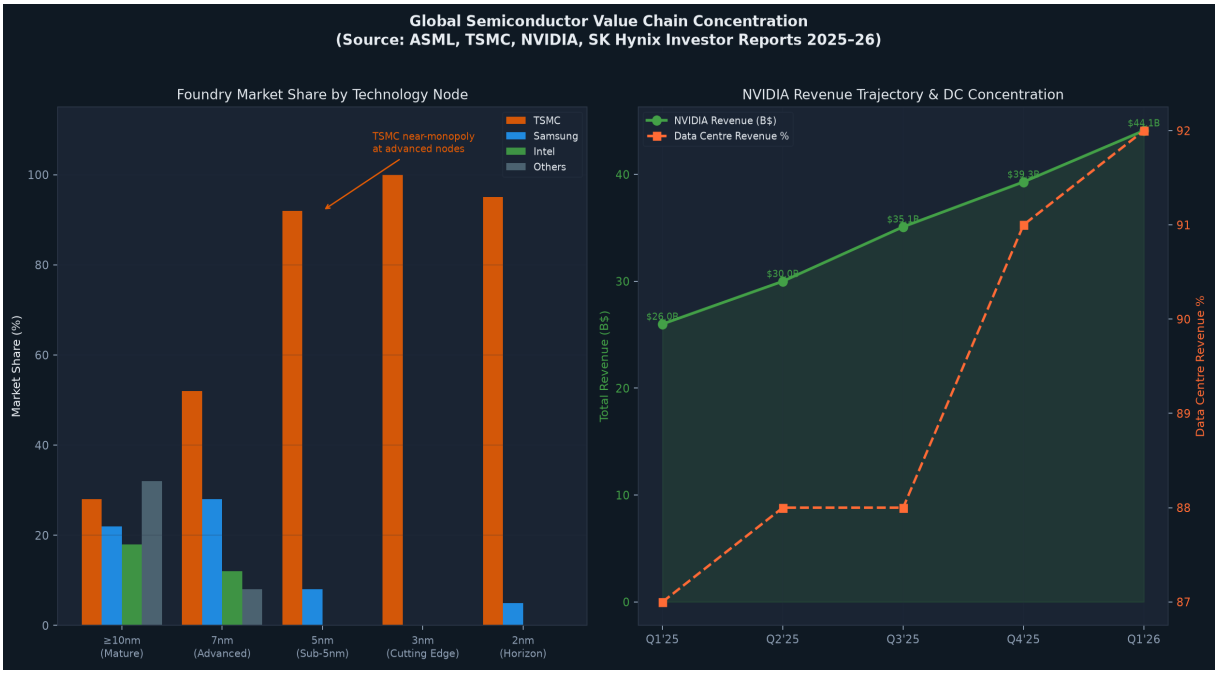


Fig. 8 — Semiconductor Value Chain Concentration. Left: TSMC holds 92% of 5nm and 100% of 3nm foundry market share—creating a single point of dependency. Right: NVIDIA data-centre revenue trajectory ($44B in Q1'26, 92% of total revenue).

VIII. India AI Infrastructure Deployment Roadmap 2024–2030

Fig. 9 synthesises milestone data from IndiaAI Mission announcements, MeitY GI Cloud tenders, and JLL pipeline reports into a unified deployment timeline. The most critical planning action is substation pre-provisioning: with 18–36 month lead times for 220kV/400kV substations, projects targeting 2028–29 commissioning must initiate grid agreements by end-2026.

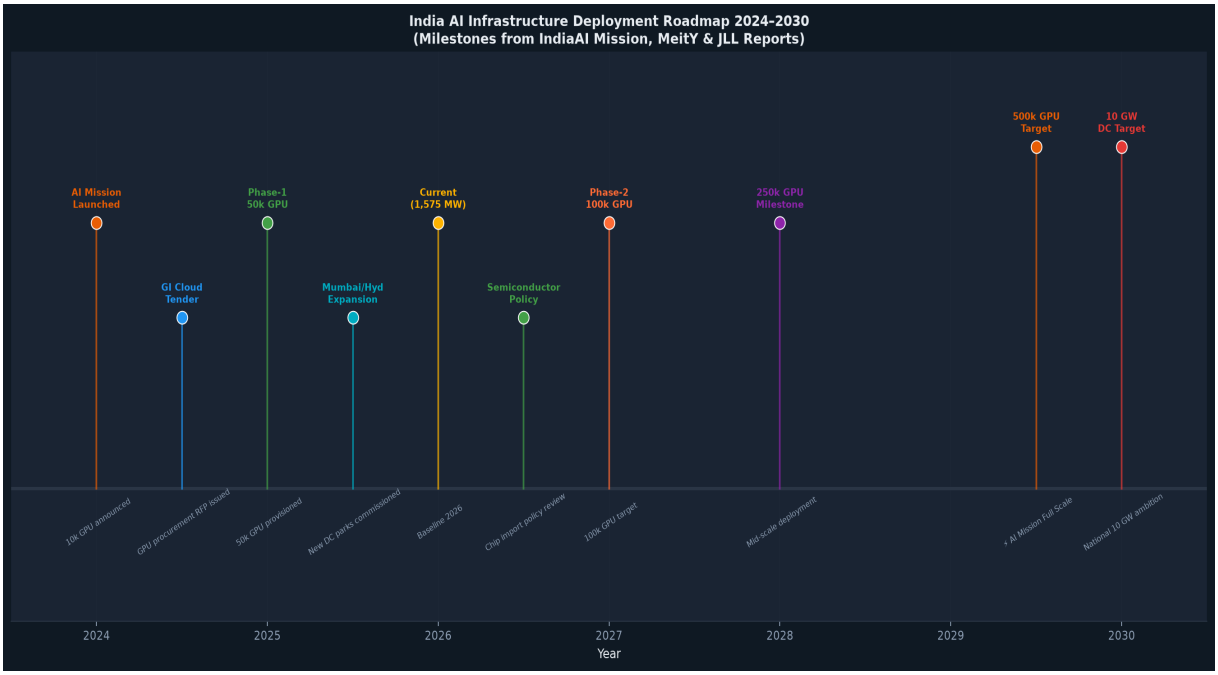


Fig. 9 — India AI Infrastructure Deployment Roadmap 2024–2030. Critical path analysis shows substation pre-provisioning must begin by Q4 2026 for 2029 targets.

IX. Policy Recommendations

R1 — Demand-Disaggregated Planning: Replace aggregate MW targets with layer-by-layer demand plans (power, water, packaging, grid) using the six-layer cascade model as a planning instrument.

R2 — Substation Pre-Provisioning: Mandate concurrent grid upgrade applications for all DC projects above 50 MW, with dedicated DISCOM fast-track approval windows.

R3 — Immersion Cooling Mandate for High-Density: All AI DC deployments exceeding 30 kW/rack should be required to use DLC or immersion cooling technologies, given tropical PUE penalties of air-cooling.

R4 — CoWoS & HBM3e Import Coordination: Establish a Strategic AI Component Reserve with bilateral agreements with TSMC (CoWoS), SK Hynix, and Micron to secure India-allocated capacity 24 months in advance.

R5 — Water Stress Siting Guidelines: Prohibit new cooling-tower-based data centres in water-stressed metro zones (WSI > 3.5); incentivise coastal or river-proximate sites with sea/river water cooling.

X. Conclusion

This paper has demonstrated that India's 500k-GPU AI Mission target is physically achievable but operationally non-trivial. The Six-Layer Demand-Cascade model reveals that the binding constraints are not semiconductor fabrication (wafer starts represent only 1.6% of TSMC output) but rather advanced packaging (CoWoS at 7.41%), high-voltage grid infrastructure (18–36 month lead times), and tropical cooling efficiency (PUE 1.40–1.65 vs 1.20–1.35 in temperate climates). India's 38,000 AI professionals represent a genuine strategic asset; translating that intellectual capital into deployed compute capacity requires demand-disaggregated planning, early supply-chain engagement, and tropical-optimised data centre design standards. The demand-cascade framework presented here provides both the analytical tool and the empirical baseline for that planning.

References

[1] JLL, "India Data Centre 2026 Mid-Year Report," JLL Research, Jul. 2026.

[2] Stanford HAI, "AI Index Report 2025," Stanford University, Apr. 2025.

[3] IndiaAI Mission, "Progress Update — AI Compute Deployment Phase 1," MeitY, Mar. 2026.

[4] ASML, "Investor Day Presentation 2025," ASML Holding N.V., Nov. 2025.

[5] TSMC, "2025 Annual Report," Taiwan Semiconductor Manufacturing Co., Mar. 2026.

[6] NVIDIA, "Q1 FY2026 Earnings Release," NVIDIA Corporation, May 2026.

[7] SK Hynix, "Q4 2025 Earnings Conference Call Transcript," SK Hynix Inc., Feb. 2026.

[8] IEA, "Electricity 2025: Analysis and Forecast to 2027," International Energy Agency, 2025.

[9] BloombergNEF, "Global Data Centre Market Outlook 2026," Bloomberg Finance L.P., Jan. 2026.
[10] CBRE, "Asia Pacific Data Centre Trends H2 2025," CBRE Group Inc., Jan. 2026.
[11] R. P. Hamlen et al., "Murphy's Yield Model for Advanced Node Wafer Processing," IEEE Trans. Semicond. Manuf., 2019.
[12] D. Tirpak et al., "Advanced Packaging Technologies for AI Accelerators," IEDM Proc., 2024.
[13] A. Shehabi et al., "United States Data Centre Energy Usage Report," LBNL, 2024.
[14] MeitY, "Digital India Data Centre Policy 2.0 Draft," Ministry of Electronics and IT, 2025.
[15] NASSCOM, "India AI Infrastructure Investment Trends 2025," NASSCOM Foundation, 2025.